

## What is a Supply Chain?

A supply chain is just the journey a product takes from being made to reaching you. Think about a simple packet of biscuits sitting in a Kirana store. That packet didn't appear out of nowhere, it came from a distributor, who got it from a warehouse, who got it from a factory, who got raw materials from a supplier. Every one of those steps is part of the supply chain.

The structure looks like this:



Each box in that chain holds some inventory — stock sitting and waiting. And that's exactly where the problem begins.

## What is Inventory and Why Does it Exist?

Inventory is stock you hold today to meet demand in the future. You hold it because there's a gap between when you can get things and when customers want them.

There are three types of inventory:

| Type               | What it is                                                                                           | Real example                                                                                   |
|--------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Cycle Stock        | The regular stock you go through in normal operations - ordered and consumed in a predictable rhythm | A supermarket orders 200 litres of milk every Monday to cover the week's sales                 |
| Safety Stock       | Extra buffer held just in case demand spikes or supply is delayed                                    | Those extra 20 litres kept 'just in case' a school group comes in on Thursday                  |
| Pipeline Inventory | Stock that is ordered and paid for but still in transit - on a truck, ship, or in a warehouse queue  | The milk that left the dairy at 4am and arrives at 8am -technically owned but not yet on shelf |

# The Three Costs

| Cost                                             | What triggers it                                                                 | Why it hurts                                                                                                                                      |
|--------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Holding Cost<br>(cost of too much)               | You ordered more than customers bought - excess stock sits unsold                | Storage space, insurance, risk of expiry or obsolescence, capital tied up. Think of ice cream melting in a freezer you're paying electricity for. |
| Shortage / Stockout Cost<br>(cost of too little) | Customers want something you don't have                                          | Lost sales, unhappy customers who go elsewhere, damage to reputation. In hospitals, a drug stockout can be life-threatening.                      |
| Ordering Cost<br>(cost of placing an order)      | Every time you place a new order, there is a fixed cost regardless of order size | Administrative work, delivery setup, supplier minimums. This is why you don't order 1 unit at a time.                                             |

## The Core Tension

Holding cost goes up when you stock more. Shortage cost goes up when you stock less. These two costs pull in opposite directions - and that tension is the entire reason inventory optimization exists as a field of study. If there were no holding cost, you'd stock infinitely. If there were no shortage cost, you'd stock nothing.

# Why Local Decisions Break the Whole System

Here's something counterintuitive that took researchers decades to properly understand: even if every single node in a supply chain makes perfectly rational decisions for itself, the system as a whole can behave terribly. This is called the **bullwhip effect**.

Imagine a retailer notices a small dip in demand - say customers bought 5% less biscuits this week. Being cautious, they order 10% less from the distributor. The distributor, now seeing a 10% drop in orders, gets nervous and orders 20% less from the warehouse. The warehouse panics and cuts their factory order by 40%. The factory now sees massive swings in demand when the actual customer demand barely moved.

## The Key Insight

Small demand fluctuations at the customer end get amplified as you move upstream. A 5% customer demand change can look like a 40% demand swing to the factory. This happens because each node makes decisions based on what it sees locally and not based on actual end-customer demand. This is why sharing information across the supply chain (and optimizing it as a whole system) is so valuable.

## What to Watch

| Video                                                                                 | Why watch it                                                                             |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <a href="https://youtube.com/watch?v=IZPO5RclZEo">youtube.com/watch?v=IZPO5RclZEo</a> | The overview of what a supply chain actually is. Watch this first, before anything else. |
| <a href="https://youtube.com/watch?v=qk15U_GSCDI">youtube.com/watch?v=qk15U_GSCDI</a> | Covers the three costs (holding, shortage, ordering) .                                   |
| <a href="https://youtube.com/watch?v=3dXCaRetzdw">youtube.com/watch?v=3dXCaRetzdw</a> | Animated explanation of the bullwhip effect.                                             |

## What to Read

### What is Inventory? - Investopedia

[investopedia.com/terms/i/inventory.asp](https://investopedia.com/terms/i/inventory.asp)

## Code

```
# Week 1 - Inventory Simulation

import numpy as np
import matplotlib.pyplot as plt

# 30 days of customer demand (units per day)
demand = [45, 52, 38, 61, 49, 55, 42, 58, 47, 50,
          63, 41, 56, 48, 53, 39, 62, 44, 57, 51,
          46, 60, 43, 54, 49, 58, 40, 55, 48, 52]

# We decide to stock this many units every day
stock = 60

inventory = []
current = stock

for d in demand:
    current = current - d          # customers buy d units
    if current < 0:
        current = 0              # can't have negative inventory
    inventory.append(current)     # record end-of-day inventory
    current = stock              # restock to 60 at start of next day

# Plot demand vs inventory
plt.figure(figsize=(10, 4))
plt.plot(demand, label='Daily Demand', color='tomato', marker='o',
         markersize=4)
plt.plot(inventory, label='End-of-day Stock', color='steelblue', marker='s',
         markersize=4)
plt.xlabel('Day')
plt.ylabel('Units')
plt.title(f'Inventory Simulation - Stock Level = {stock}')
```

```
plt.legend()  
plt.grid(True, alpha=0.3)  
plt.show()
```

## Experiments

### Experiment 1 : Stock too low: change stock = 60 to stock = 35

Run the code and look at the chart. On many days the inventory line will hit zero : those are stockout days. The inventory line will look jagged and spend a lot of time at the bottom.

Ask yourself:

- How many days did you run out of stock?
- What happened to customers on those days?
- Which cost is hurting you most - holding or shortage?

### Experiment 2 : Stock too high: change stock = 60 to stock = 120

Run again. Now the inventory line will barely drop - you always have plenty. No stockouts. But look at how much excess inventory you're carrying every single day.

Ask yourself:

- Did you have any stockouts?
- How much inventory is left over at the end of each day?
- You're paying holding cost on all that leftover stock. Is that a good use of money?

### Experiment 3 : Find your own number

Try stock = 50, then 55, then 65, then 70. For each one, count stockout days manually from the chart. Try to find the value of stock that gives you the fewest stockouts without wasting too much on excess inventory.

There's no single right answer here.

# Deliverable

## Q1 -What happens at the extremes?

Run your simulation with stock = 35 and with stock = 120. For each case, describe what the chart looks like. Which cost (holding or shortage) dominates each case?

## Q2 -Is the best stock level just the average demand?

Look at the demand list, the average daily demand is about 50 units. Try setting stock = 50. Does that give you zero stockouts? If not, why not?

## Q3 - What would make this decision easier in real life?

In your simulation, you knew the demand list in advance as you could see all 30 days before the loop ran. In real life, you don't know tomorrow's demand when you're deciding how much to stock today. What kind of information would help a real retailer make a better stocking decision?

It is an open ended question try to think creatively.

Other than this feel free to add any other interesting observations you notice from the simulation as well .. maybe something surprising in the graphs, patterns in demand, how inventory behaves at different stock levels, or any real-life connections you can think of.